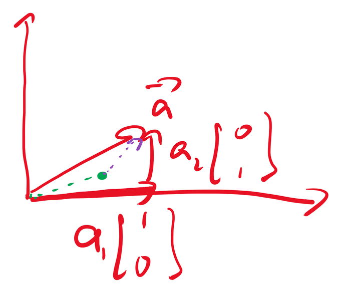
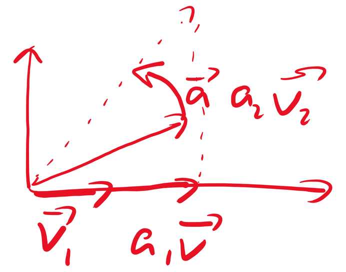


$a_1\begin{bmatrix}1\\0\end{bmatrix} + a_2\begin{bmatrix}0\\1\end{bmatrix} = b_1\begin{bmatrix}\ \end{bmatrix} + b_2\begin{bmatrix}\ \end{bmatrix}$



$W\vec{a}$

$W = V \Sigma V^{-1}$

$W\vec{a} = V \overset{(n)}{\Sigma} \underline{V^{-1}\vec{a}}$

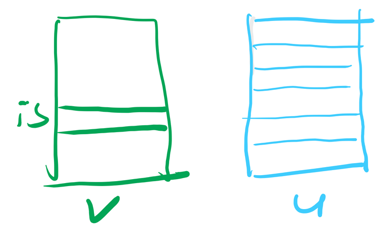
$W^T W \to U$

$W W^T \to V$

$V\vec{a}^{-1} = \begin{bmatrix}\vec{v}_1 & \vec{v}_2 & \dots\end{bmatrix}\begin{bmatrix}a_1\\a_2\\\vdots\end{bmatrix} = a_1 \cdot \vec{v}_1 + a_2 \vec{v}_2$

word2vec

cute cad is on a mat



$(001..0) = idx \leftrightarrow$

is	cute
is	cat
is	on
is	mat

||

SKIP-GRAM →

→ (001..0)	(00...1..0
(001..0	

CBOW

(01011..0)	(001...0)

$(0\,0 \dots 1 \dots 0)^T \begin{bmatrix}\ \end{bmatrix} = \vec{v}_{is}$; $\vec{v}_{is}$ … U^T

← исходные вектора OHE - строка

$Obj = -\sum_{i=0}^{|V|} p_i \log(q_i) = -1 \cdot \log(q_{cute})$

$\begin{bmatrix}(0\,0\,1\,0\,0) \\ (0\,0\,1\,0\,0 \\ \vdots\end{bmatrix} V U^T = \begin{bmatrix}(\ \sim\ ) \\ (\ \sim\ ) \\ (\ \sim\ ) \\ (\ \sim\ )\end{bmatrix}$ vs. $\begin{bmatrix}(OHE_{cat}) \\ (OHE_{cute}) \\ \dots\end{bmatrix}$

CE

← альтернативно, если вектора OHE - столбцы

$V^T \begin{bmatrix}0\\0\\\vdots\\1\\0\end{bmatrix} = \vec{v}_{is}$

$U \begin{bmatrix}\vec{v}_{is}\end{bmatrix} = \begin{bmatrix}\langle \vec{v}_{is}, \vec{u}_0 \rangle \\ \langle \vec{v}_{is}, \vec{u}_1 \rangle \\ \dots\end{bmatrix} = \begin{bmatrix}a_1\\a_2\\a_3\\\dots\end{bmatrix} \to \mathrm{softmax}\left(\begin{bmatrix}a_1\\a_2\\\dots\end{bmatrix}\right) = \vec{q} = \begin{bmatrix}q_1\\q_2\\\dots\end{bmatrix}$ vs. $\underset{p}{\begin{bmatrix}0\\0\\\vdots\\1\\\dots\end{bmatrix}}$

$\vec{a}$

$(U V^T [OHE_{is}])^T = (OHE_{is})^T (U V^T)^T =$

$= (OHE_{is})^T V U^T$